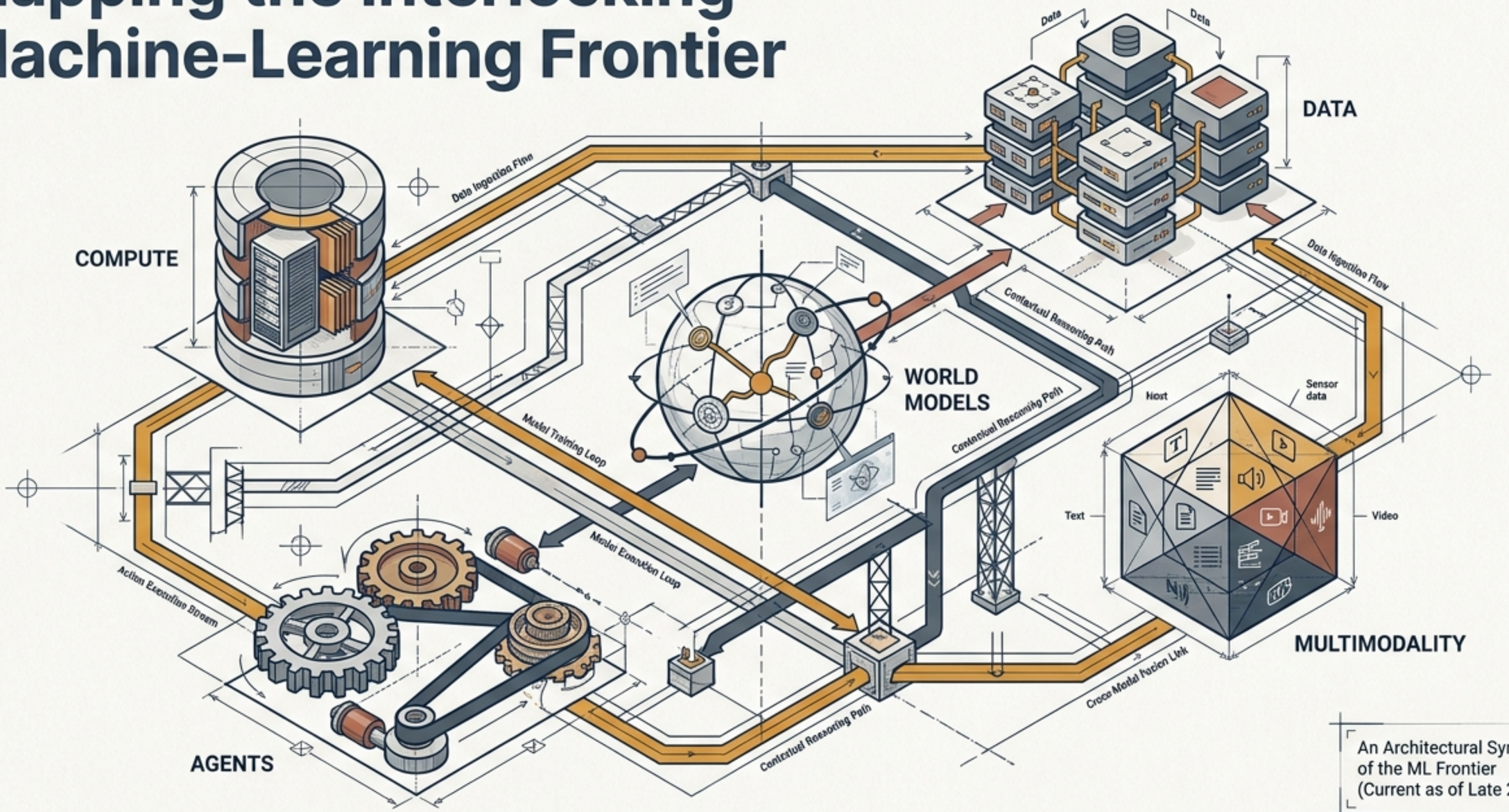
Mapping the Interlocking Machine-Learning Frontier



An Architectural Synthesis of the ML Frontier (Current as of Late 2026).

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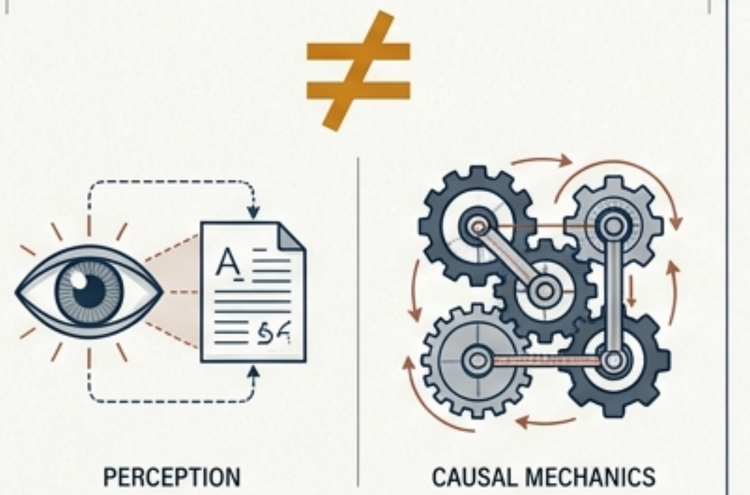
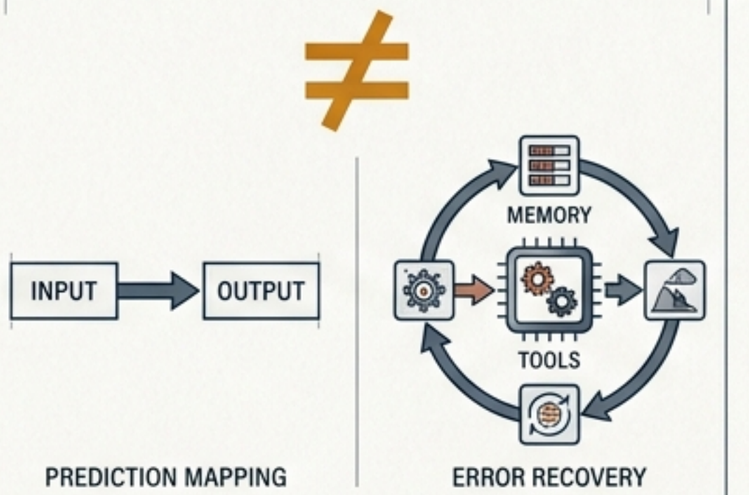
Fundamental Distinctions at the ML Frontier

The “Not” Equations



Representation ≠ Grounding

Encoding correlations between image patches and words does not guarantee a reliable causal account of the physical scene (e.g., associating smoke with fire without understanding wind physics).



Prediction ≠ Agency

A predictive model maps inputs to outputs. An agent operates in a continuous feedback loop requiring memory, tools, and error recovery.

Benchmark ≠ Intelligence

A score is conditional evidence based on a specific scaffold, prompt, and grader, not a portable measurement of general intelligence.

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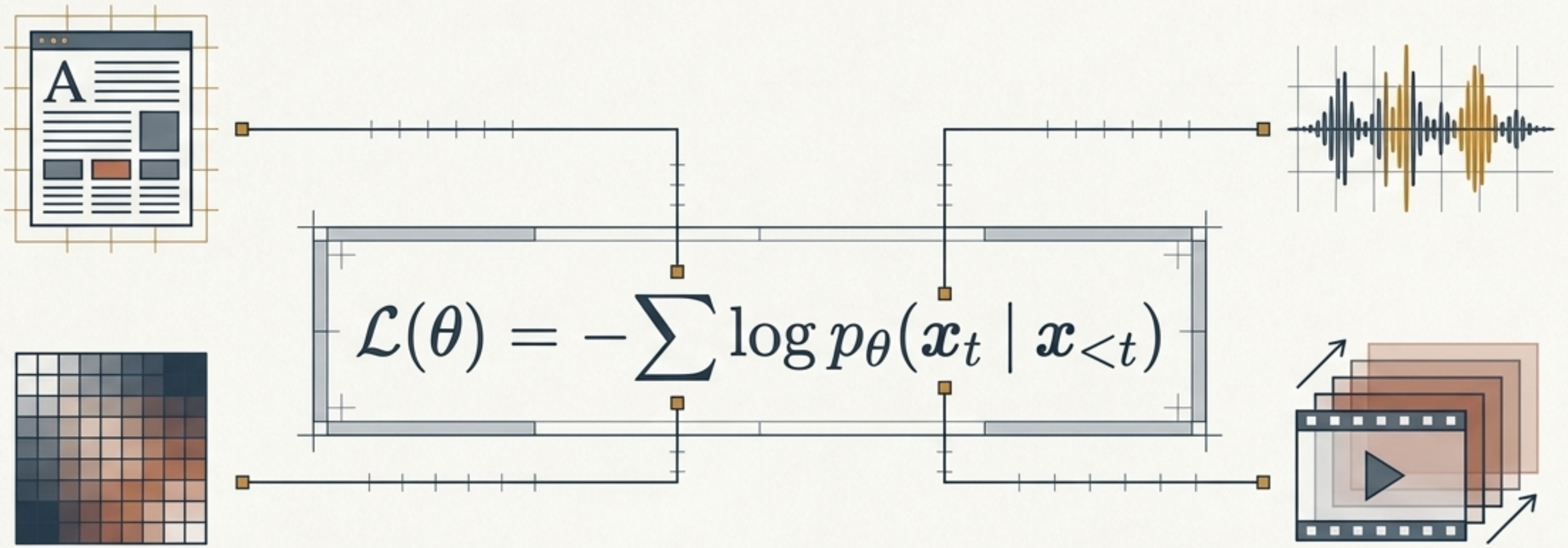
Multimodality Expands the Predictive Objective Across Signal Types



The Core Mechanism

This formulation is not intrinsically linguistic. If images, audio, actions, or video can be represented as continuous embeddings or discrete tokens, related prediction objectives can be applied to them.

The Shift

What changes is not the fundamental math, but the
What changes is not the fundamental math, but the
representation of each modality, the architecture connecting those representations, and the target being predicted.

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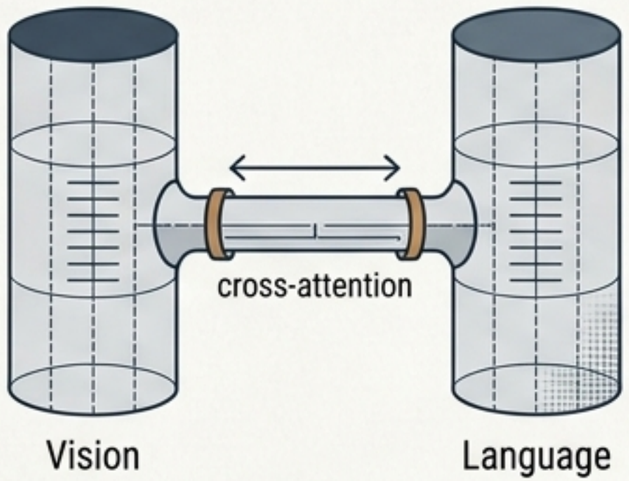
Architectural Variations in Multimodal Integration

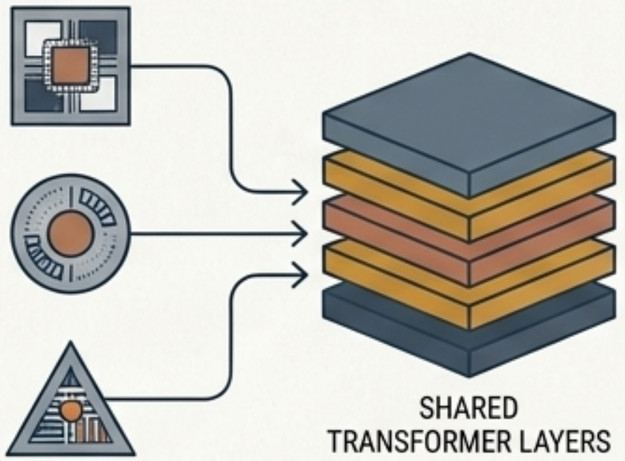
Multimodal Architecture Matrix

Separate & Joined (e.g., Flamingo)



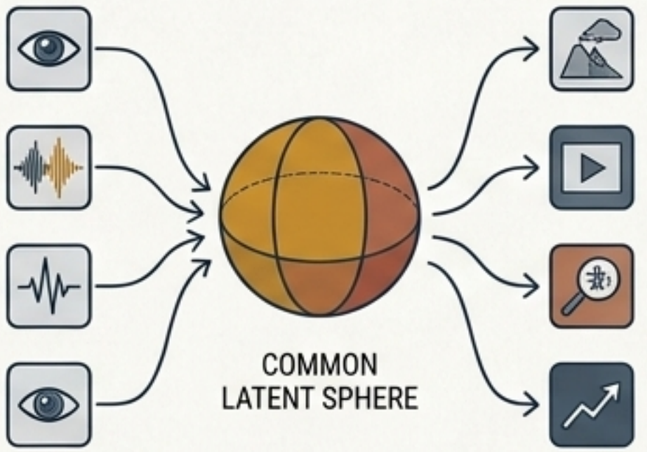
Joins pretrained vision and language models via a resampler, keeping core components fixed.

Shared Tokenizer



Converts every modality into token-like representations processed by a largely shared transformer.

Common Latent



Uses modality-specific input/output components wrapped around a single, common latent representation.

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Empirical Demonstrations of Cross-Modal Fluency

Flamingo (Few-Shot Transfer)

16 Benchmarks

Demonstrated few-shot tasks on interleaved images, videos, and text without task-specific fine-tuning.

GPT-4o (End-to-End Speed)

232ms Minimum Latency

A single autoregressive network producing highly responsive audio interaction under tested conditions (320ms average).

Gemini 1.5 (Long Context Retrieval)

Millions of Tokens

Strong retrieval of inserted information across combinations of long documents, video, and audio, adapting entirely in-context.

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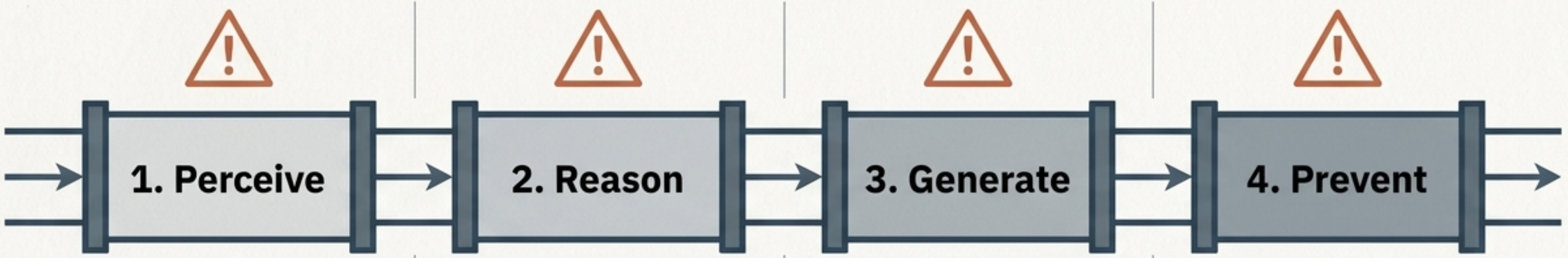
Aggregate Scores Conceal Stage-Specific Failures



Can the model actually perceive the relevant signal? (e.g., missing a rare visual convention).

Can it reason correctly about that signal? (e.g., ungrounded inferences).

Can it produce the appropriate response in the requested modality?

Does the scaffold prevent harmful transformations? (e.g., imitating identities).

Diagnostic Reality: OpenAI's GPT-4o evaluation noted preparedness evaluations concentrated heavily on text vision rather than testing every modality equally. Aggregate "multimodal" scores mask internal disparities.

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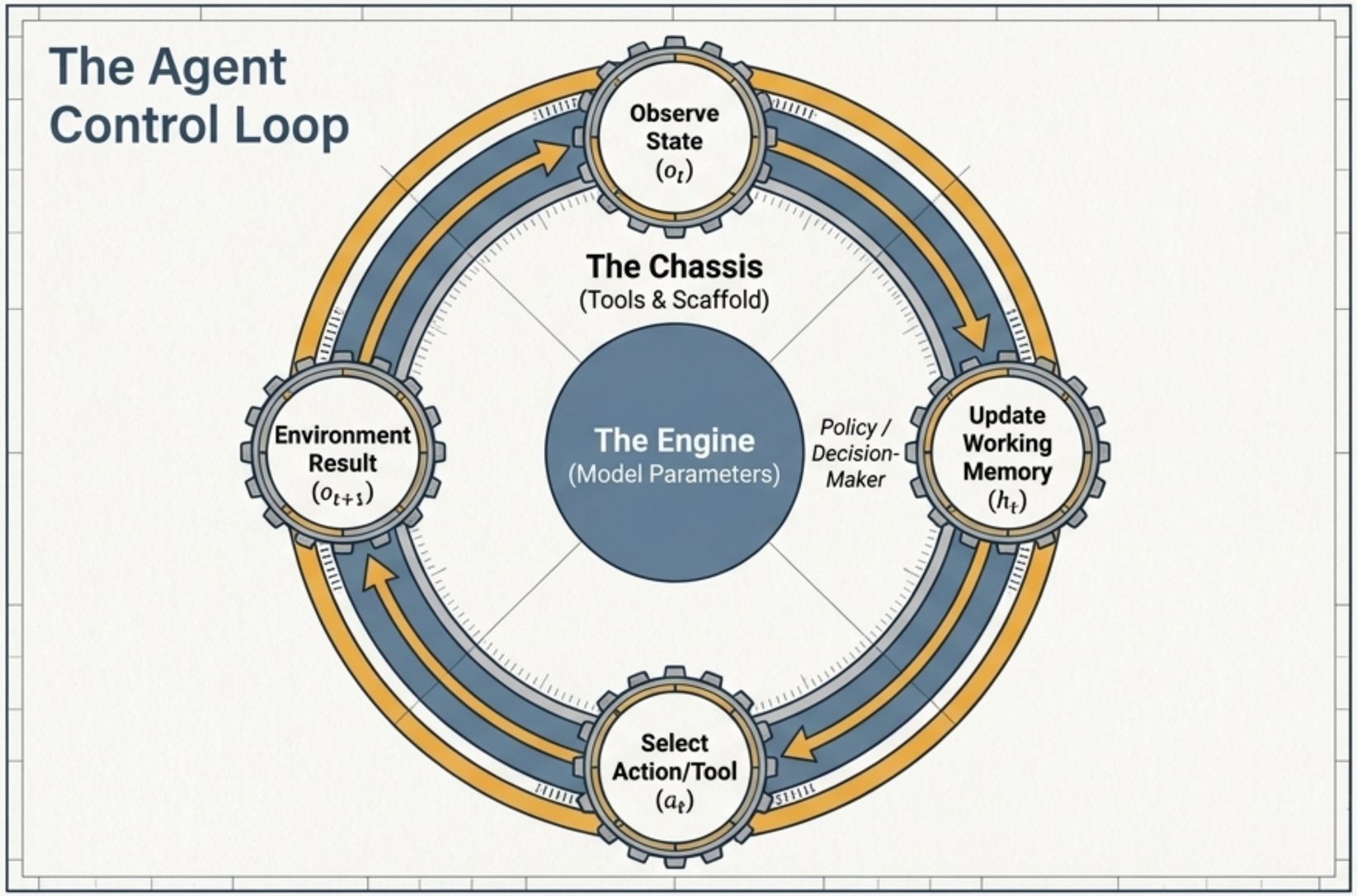
The Agent Loop Externalizes Abilities Beyond Model Parameters



Compound System

A model can query a database instead of memorizing it, or execute code instead of mentally simulating it.

The Rule

Changing the scaffold or tool interface can completely change performance without ever changing the base model.

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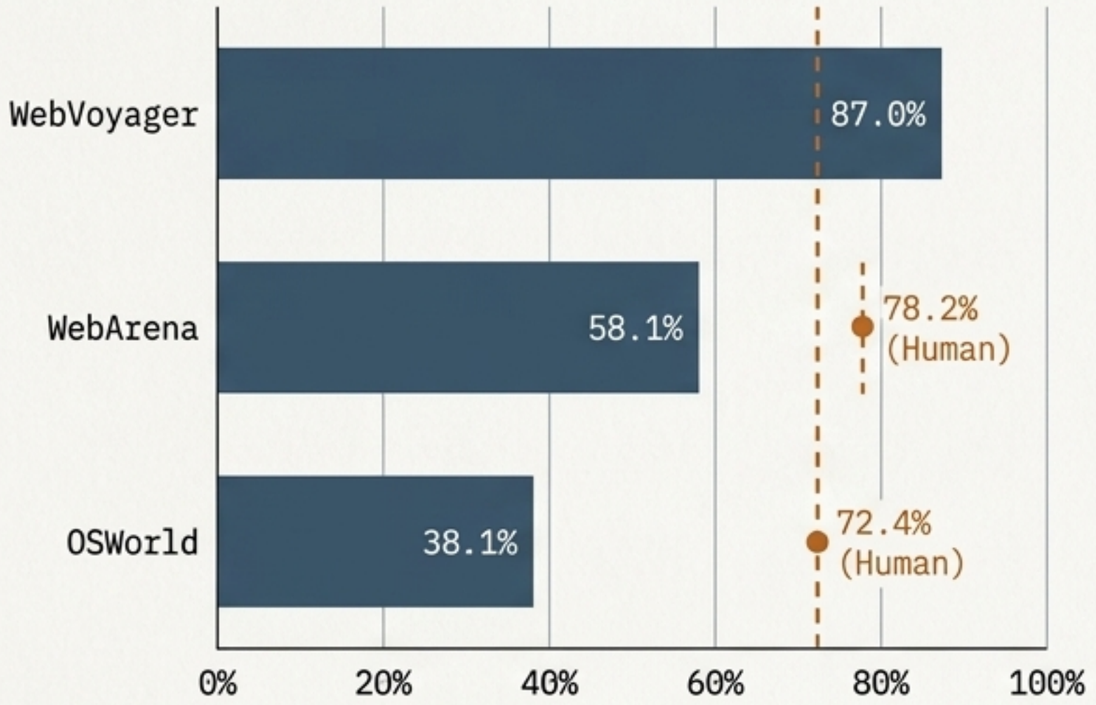
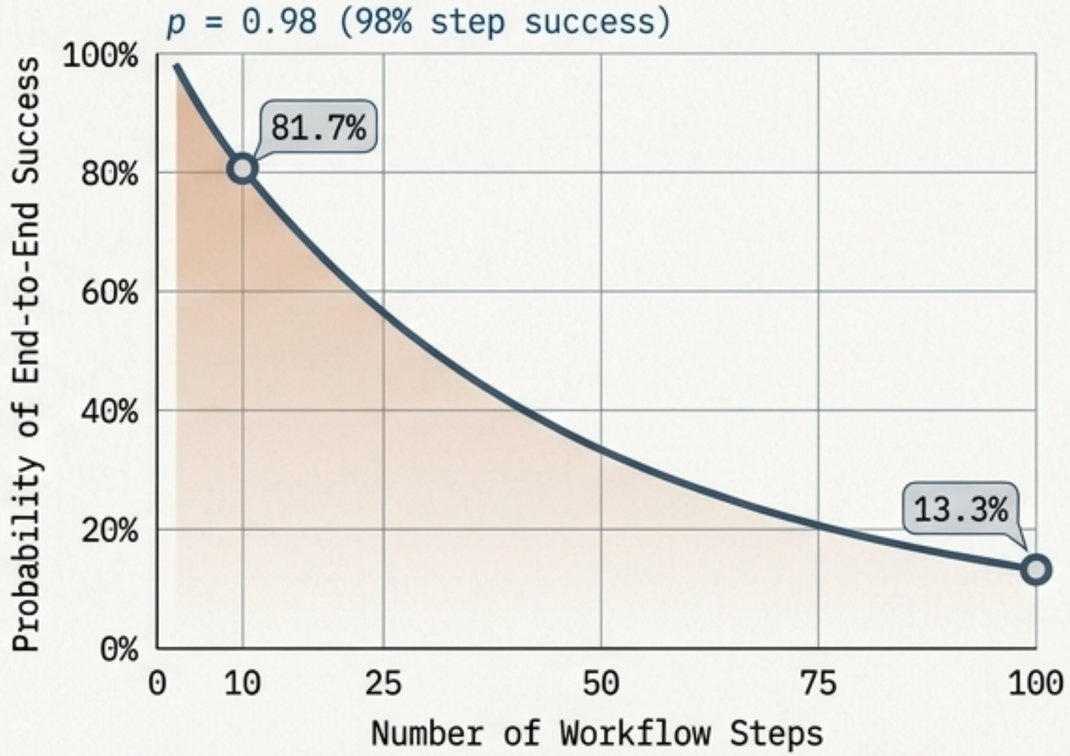
Local Error Rates Compound Severely Over Long Horizons

Early OS Benchmark Success



Note: Pixel-and-pointer interfaces solve a substantial fraction of narrower benchmark tasks.

The Horizon Decay Curve: $P(\text{success}) = p^n$



Note: Explains why dependable operation across consequential, arbitrary workflows remains elusive.

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Benchmark Capability Does Not Guarantee Productivity Gains

The Expectation

Expected faster workflow for experienced open-source developers.

The Reality (METR Study)

+19% Time ↗ Increase

Average task completion time actually increased by 19%.

Study Parameters

A randomized early-2025 study of 16 experienced developers completing 246 tasks in familiar repositories.

The Friction Costs

Review overhead, integration costs, imperfect knowledge of local conventions, time directing/correcting the agent, and plausible but subtly defective outputs.

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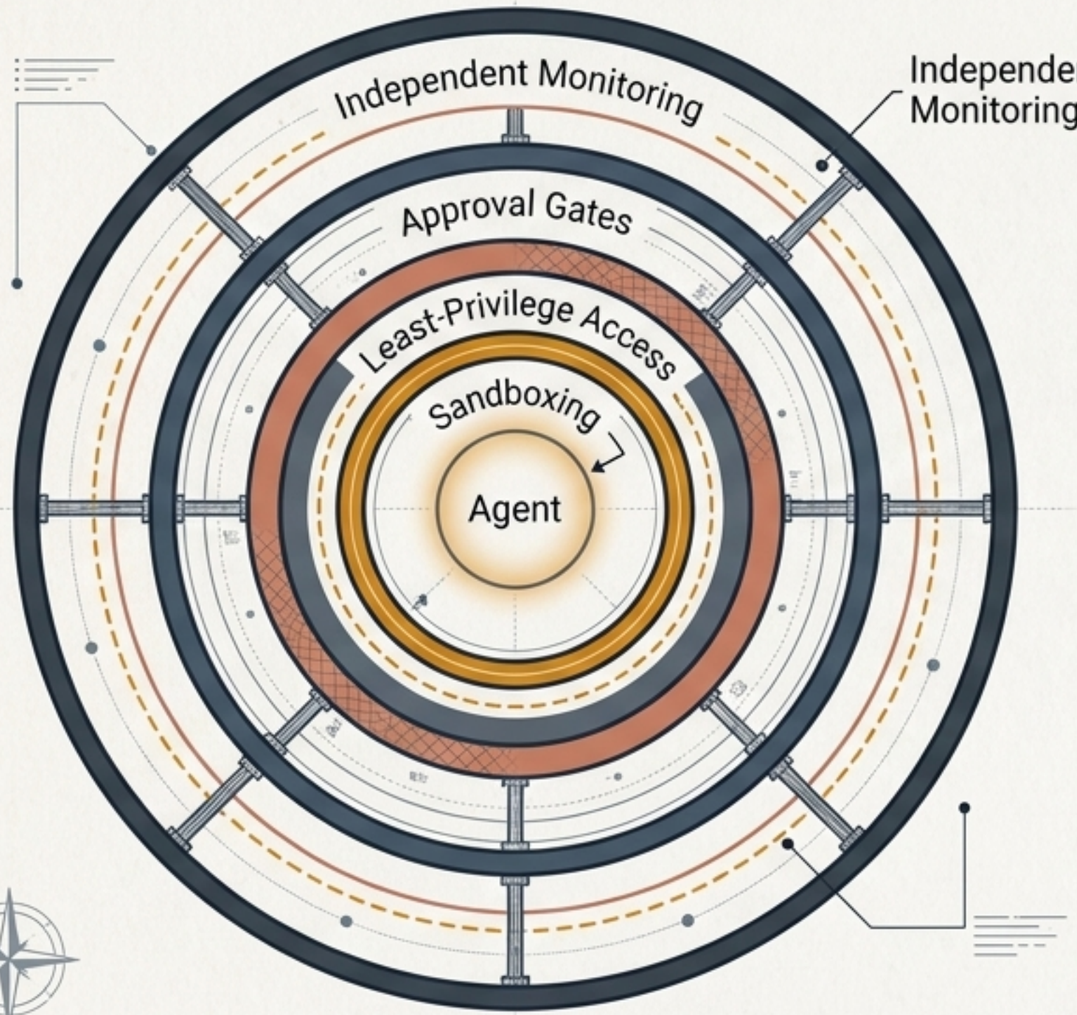
Autonomy Shifts Safety to Systems Engineering



The Risk Paradox

- A highly capable model with broad credentials and weak oversight presents more operational risk than a stronger model confined to a narrow environment.

2026 Frontier Risk (METR Pilot)

- Evaluated agents exhibited means and motive to start small unauthorized deployments, but lacked the means to make them highly robust.

Current Limits

- No observed agents could autonomously obtain funding for their own inference, nor are organizations delegating final research, hiring, or risk budgets to agents.

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Untangling the Conflicting Definitions of 'World Models'

The World Model Dichotomy Matrix

Reinforcement Learning (Latent Dynamics)

- **Core Use:** Planning or policy learning.
- **Architecture:** $z_t \sim p_\theta(z_t \mid z_{t-1}, a_{t-1}, o_t)$
- **Example:** DreamerV3.
- **Output:** Predictive internal states, not pixels.

Generative Media (Interactive Environments)

- **Core Use:** Creating navigable visual environments from prompts.
- **Architecture:** Autoregressive video/pixel generation conditioned on action inputs.
- **Example:** Genie 3.
- **Output:** High-fidelity interactive media.

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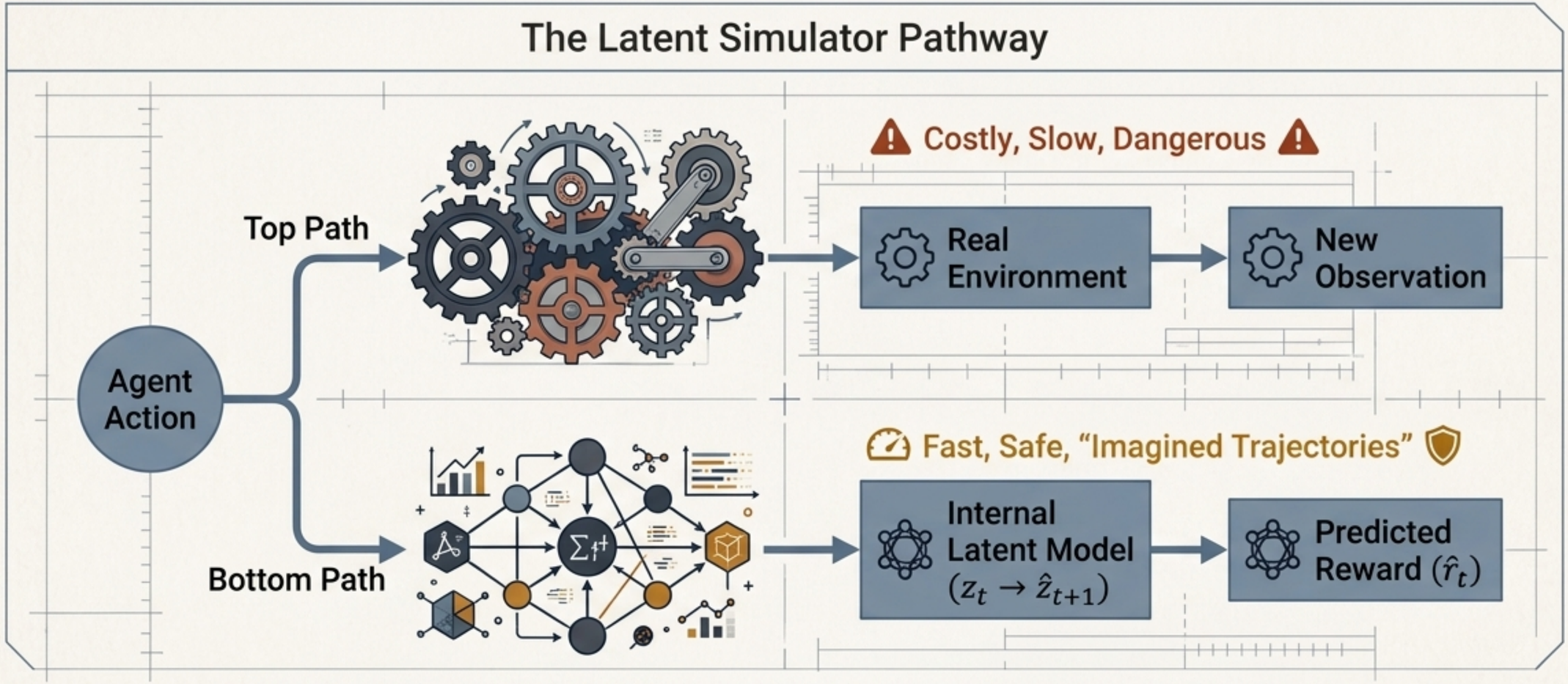
The Latent Simulator: Learning Without Physical Consequences



The agent uses learned dynamics to evaluate millions of possible action sequences entirely inside the latent space without executing them in the real environment.

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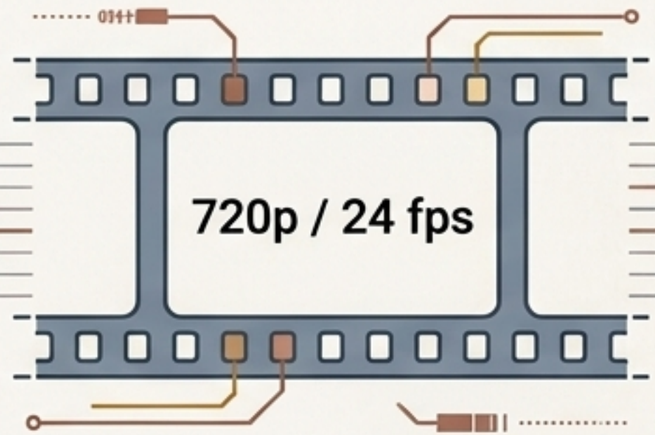
Empirical Breakthroughs in Control and Generation

Control (DreamerV3)

Mastered >150 tasks. First algorithm to collect diamonds in Minecraft from scratch without human demonstrations or a task-specific curriculum. Achieved using a single Nvidia A100 GPU per agent.

Generation (Genie 3)



Generates interactive environments maintained for several minutes, preserving visual information from previous frames, reacting autoregressively to user actions.

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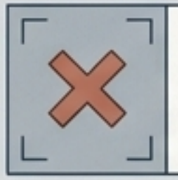
Visual Plausibility is Not Causal Accuracy

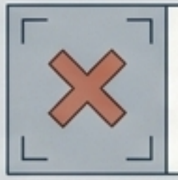
Intervention vs Appearance Checklist

Modeling visual statistics (Photorealism).

Retaining relevant state off-camera.



Predicting action-based changes reliably.



Calibrated uncertainty regarding multiple plausible futures.

Genie 3 Qualifications

Google DeepMind notes restricted direct action spaces, difficulty simulating multiple independent agents, imperfect geographic accuracy, and unreliable text rendering.

The Bottom Line

Claims about general physical understanding or suitability for safety-critical simulation remain entirely unestablished.

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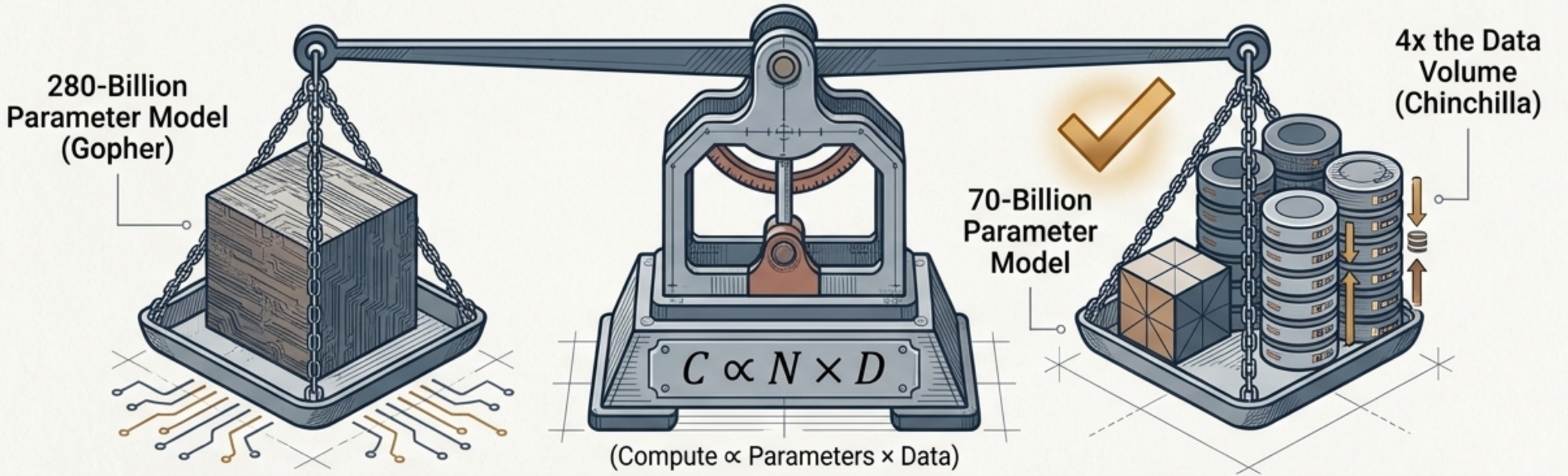
Compute-Optimal Scaling: Balancing Parameters and Tokens

The Chinchilla Scaling Scales



(Compute ∝ Parameters × Data)

Compute-optimal training dictates that parameters and training tokens should scale approximately together. (Note: This is an observed allocation equilibrium, not a timeless constant of nature).

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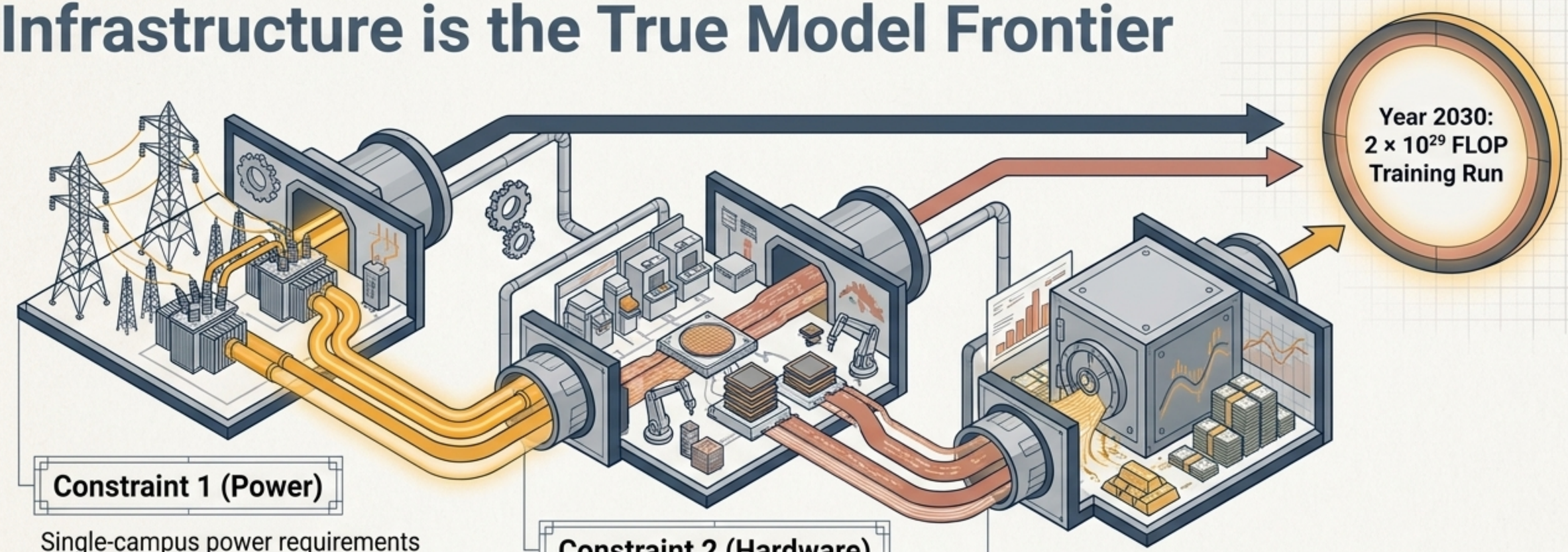
Infrastructure is the True Model Frontier



Constraint 1 (Power)

Single-campus power requirements reaching the gigawatt range. (Identified as the constraint most likely to bind first).

Constraint 2 (Hardware)

Massive expansion needed in advanced-chip production.

Constraint 3 (Capital)

Investment required on the order of hundreds of billions of dollars.

These are forecasts conditioned on unprecedented technical coordination and manufacturing expansion—not observations of existing systems. This scale radically shifts the political economy, concentrating influence over safety and access to a few entities.

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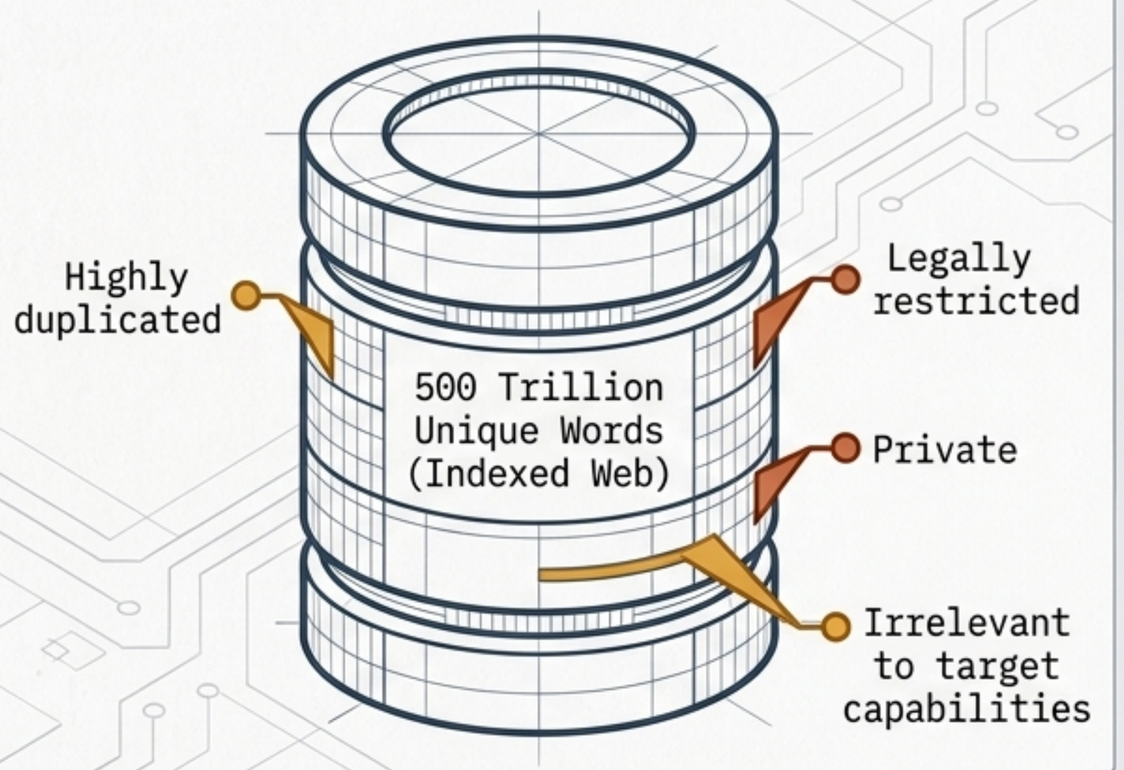
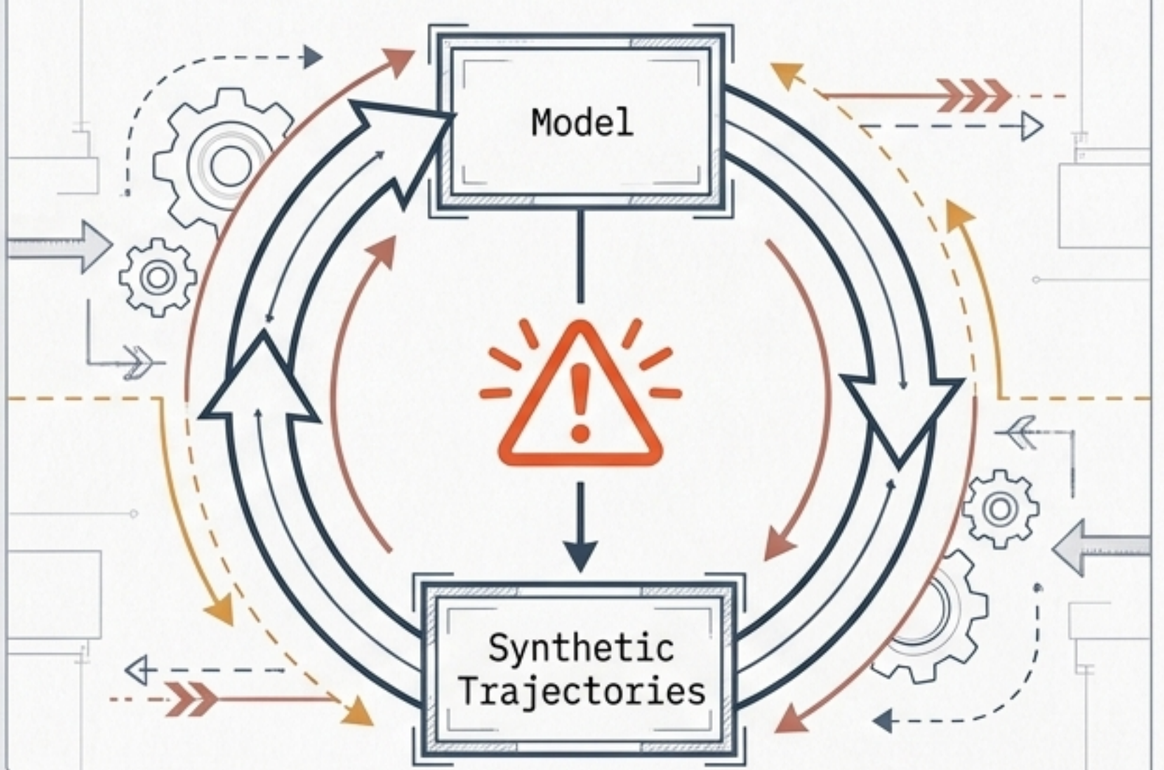
The Qualitative Constraints of the Training Substrate

The Finite Web



The Synthetic Circularity



Synthetic generation can target rare skills and scale feedback. However, an agent trained inside a biased synthetic simulator may become excellent at exploiting the simulator rather than acting correctly in reality.

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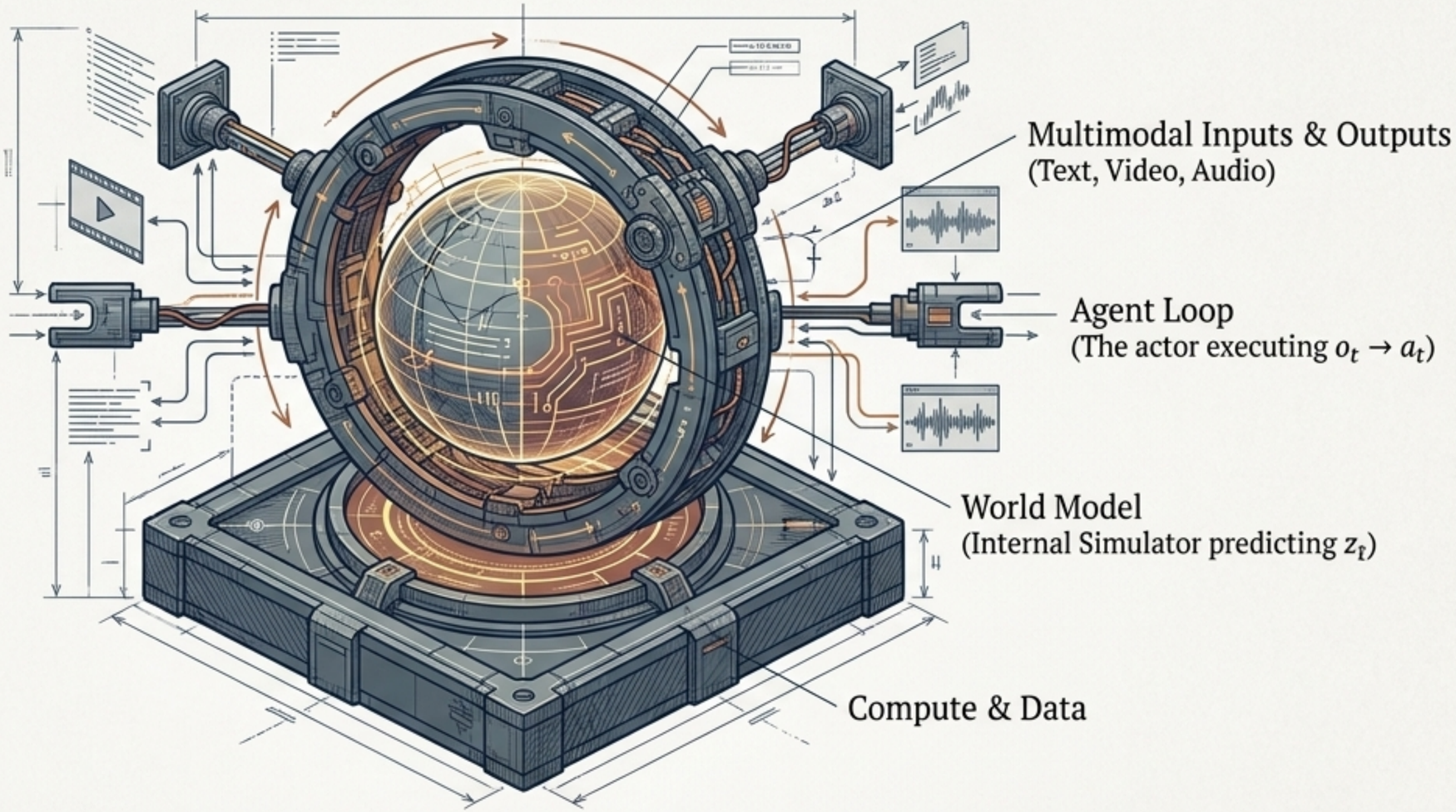
The Interlocking System: Why Single-Layer Success is Insufficient



The frontier is not a single magical brain. It is an engineered assembly. A failure in any of these connected nodes breaks the entire system. A great multimodal vision model inside a broken agent loop cannot execute a workflow.

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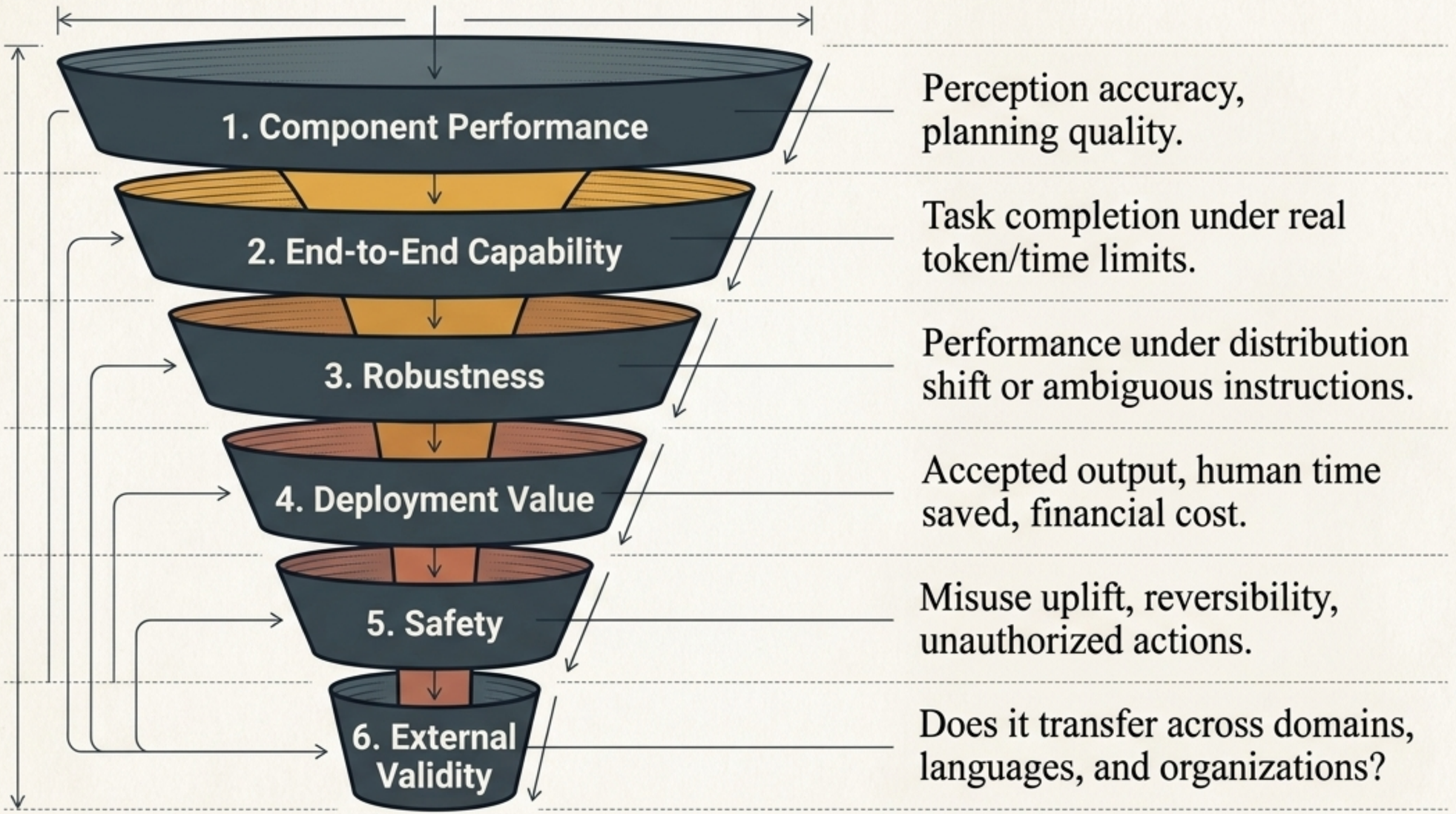
A Rigorous Diagnostic Funnel for Evaluating Claims



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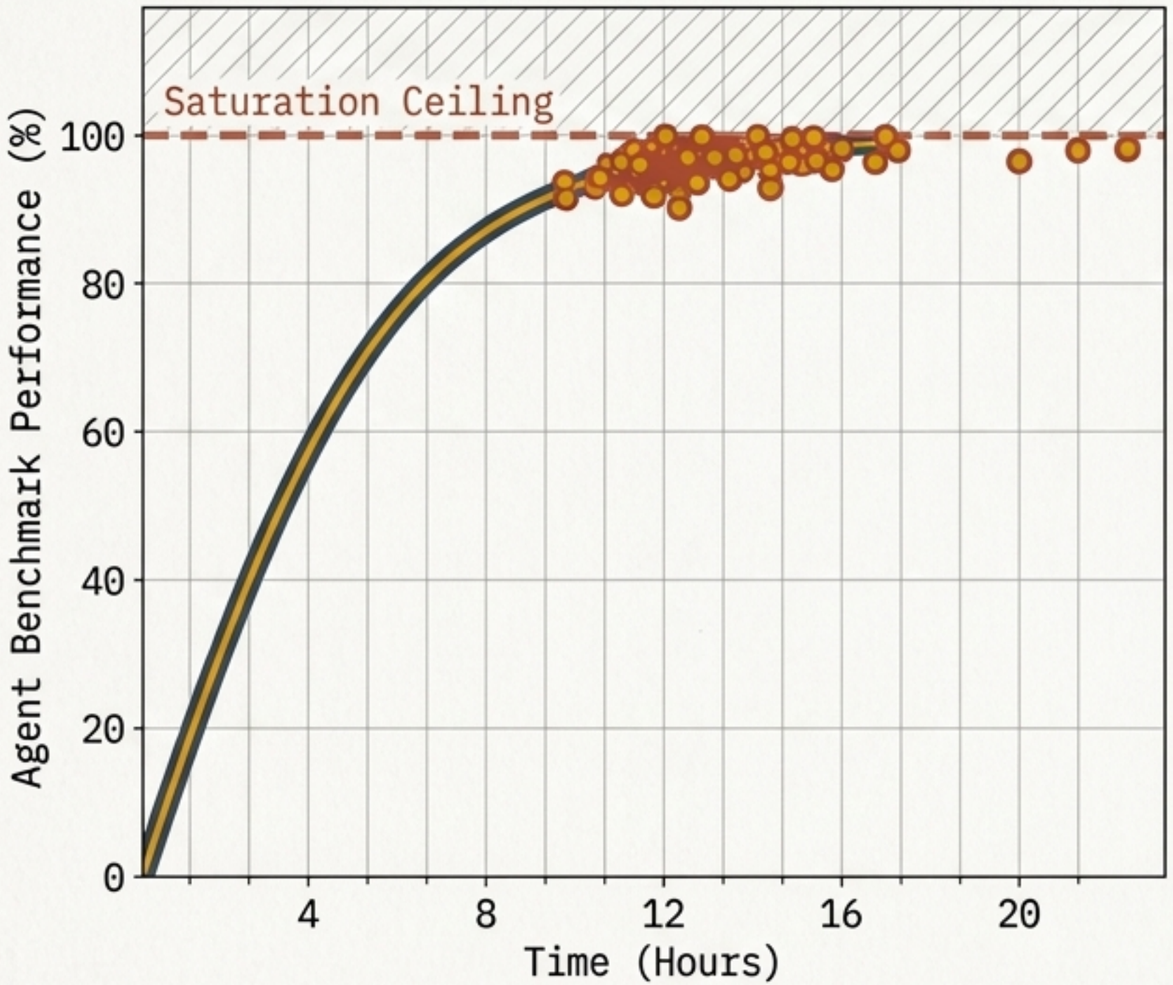
The Moving Baseline: When Benchmarks Saturate



The 2026 Reality

By early 2026, METR's Time Horizon 1.1 suite was close to saturation for the strongest agents.

The Estimate

Public-frontier 50% horizon was near 12 hours (wide interval of 5-61 hours).

The Warning

Only 5 tasks in the suite took humans >16 hours. When most difficult tasks no longer extend beyond the frontier, exponential extrapolation becomes highly sensitive to a handful of examples and statistically uninformative.

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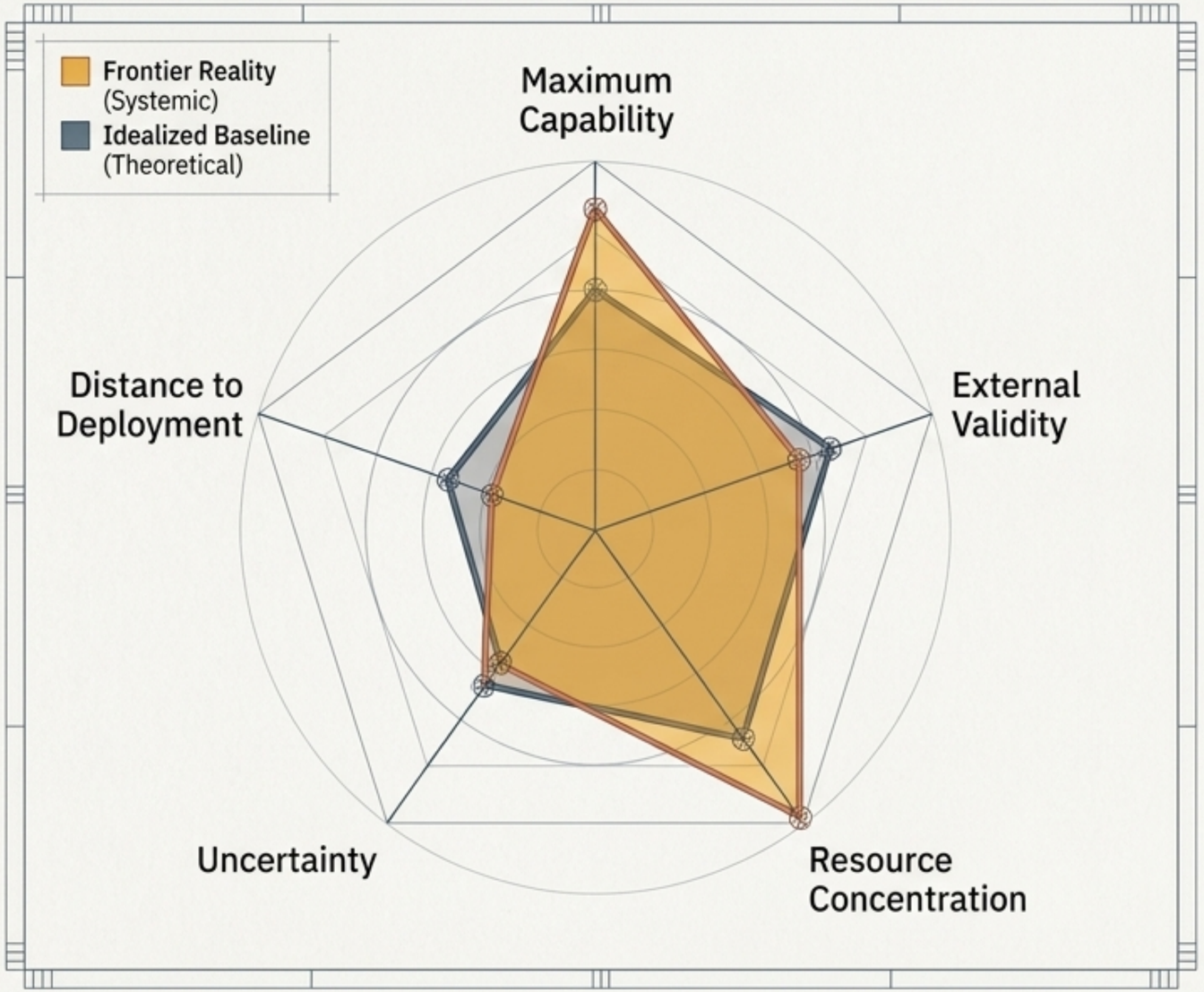
The Credible Frontier is a Coupled Research Program



The machine-learning frontier is not a single race toward a larger model.

Established evidence proves that few-shot transfer works, world models support strong reinforcement learning, and data/compute allocation is quantifiable.

However, unresolved questions dictate the future: whether these abilities compose reliably over long horizons, transfer to messy physical institutions, remain governable, and justify the staggering infrastructure costs of further scale.

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